

Infrastructure Provisioning

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Sevima, sebuah perusahaan yang bergerak di bidang teknologi informasi, memiliki rencana besar untuk mengembangkan infrastuktur mereka dengan membangun sebuah pusat data (data center). Pusat data ini diharapkan mampu menunjang performa aplikasi serta berfungsi sebagai Cadangan (backup) sehingga jika terjadi kendala, recovery dapat dilakukan dengan cepat dan efisien. Anda, sebagai calon System Administrator, diberi tugas penting untuk merancang dan mengimplementasikan topologi jaringan untuk pusat data ini menggunakan Cisco Packet Tracer.

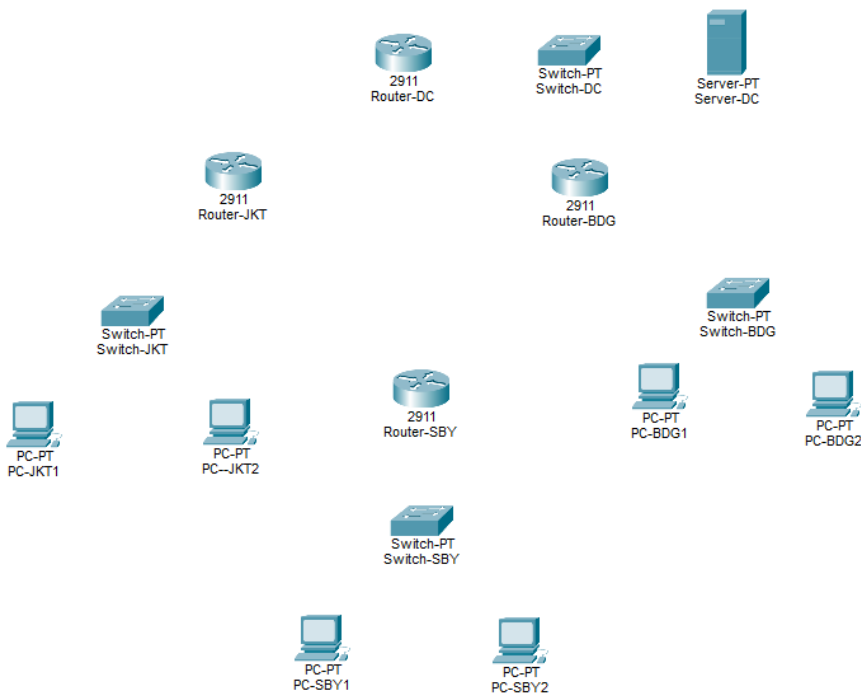
Tugas Anda tidak hanya berhenti di situ. Sevima juga berencana untuk memperluas jangkauan operasional mereka dengan membuka cabang di tiga kota besar di Indonesia: Jakarta, Bandung, dan Surabaya. Anda harus memastikan bahwa jaringan di ketiga cabang tersebut terhubung satu sama lain dengan baik.

Konfigurasi

Step 1 – Tambahkan Device

Berikut merupakan spesifikasi hardware yang dibutuhkan:

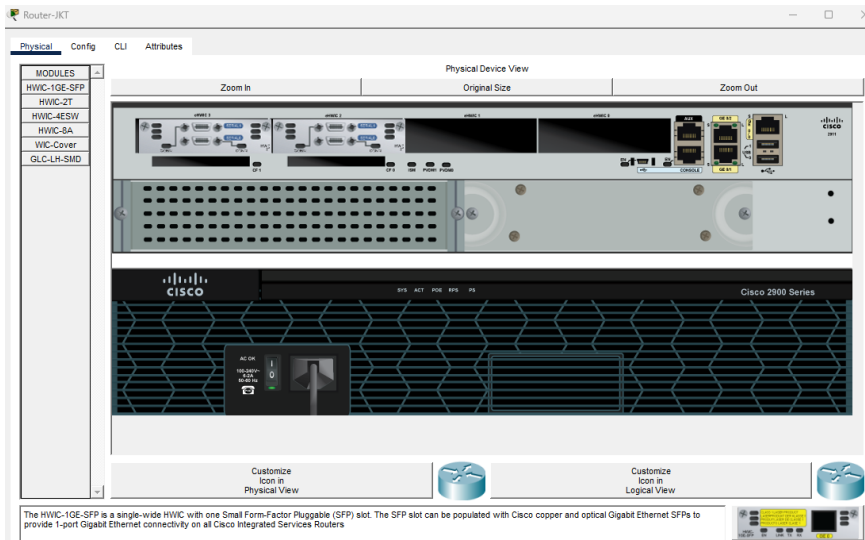
Device	Model	Qty	Tujuan
Router	2911	4	Router-DC, Router-JKT, Router-BDG, Router-SBY
Switch	2960	4	SW-JKT, SW-BDG, SW-SBY
PC	Generic	6	2 per branch (PC-JKT1/2, PC-BDG1/2, PC-SBY1/2)
Server	Generic	1	Server data center



Step 2 – Tambahkan modul serial WAN ke Router

Tiap router perlu modul **WIC-2T** untuk koneksi WAN

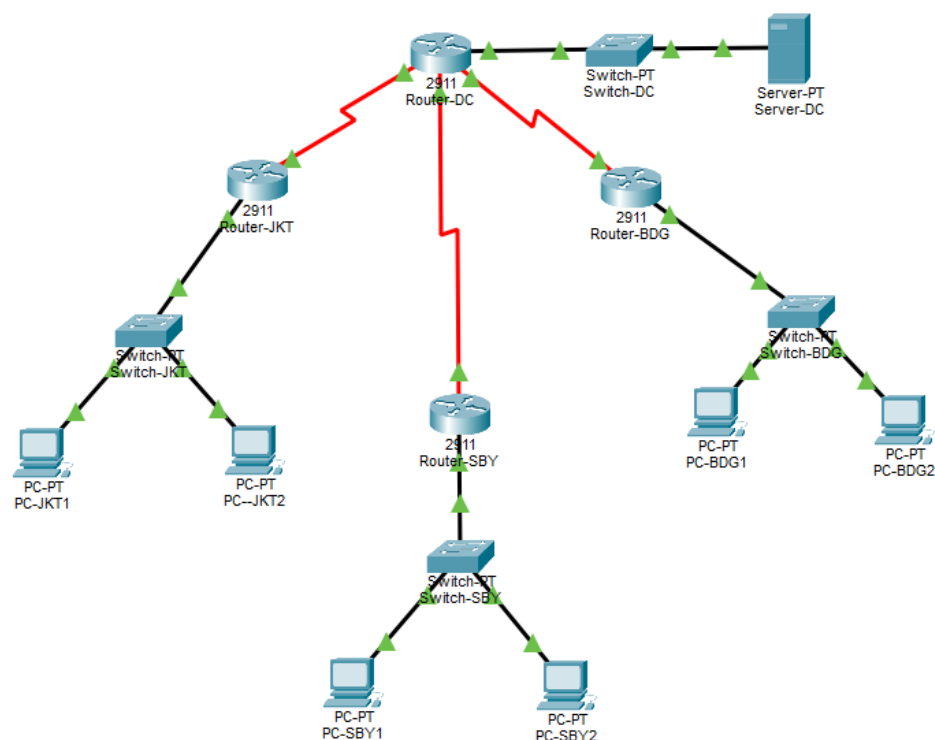
1. Klik router → ke tab Physical
2. Matikan / power off
3. Tambahkan modul **WIC-2T** ke slot kosong
4. Nyalakan Kembali / power on
5. Lakukan untuk semua Router yang tersedia



Step 3 – Koneksi Antar-Device

Spesifikasi jenis kabel yang digunakan:

Device	Jenis Kabel
Router-Router	Serial DCE/DTE
Router-Switch; Switch-PC	Copper Straight-Through



Step 4 – IP Addressing

Berikut merupakan tabel perencanaan IP addressing:

Link	Subnet	IP DC	IP Branch
DC ↔ JKT	10.0.12.0/30	10.0.12.1	10.0.12.2
DC ↔ BDG	10.0.13.0/30	10.0.13.1	10.0.13.2
DC ↔ SBY	10.0.14.0/30	10.0.14.1	10.0.14.2

Berikut merupakan konfigurasi CLI tiap Router:

1. Router-DC

```
Router>enable
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname Router-DC
Router-DC(config)#int Gi0/0
Router-DC(config-if)# ip address 192.168.100.1 255.255.255.0
Router-DC(config-if)# no shutdown

Router-DC(config-if)#int Se0/3/0
Router-DC(config-if)# ip address 10.0.12.1 255.255.255.252
Router-DC(config-if)# clock rate 64000
Router-DC(config-if)# no shutdown

%LINK-5-CHANGED: Interface Serial0/3/0, changed state to down
Router-DC(config-if)#int Se0/3/1
Router-DC(config-if)# ip address 10.0.13.1 255.255.255.252
Router-DC(config-if)# clock rate 64000
Router-DC(config-if)# no shutdown

%LINK-5-CHANGED: Interface Serial0/3/1, changed state to down
Router-DC(config-if)#int Se0/2/0
Router-DC(config-if)# ip address 10.0.14.1 255.255.255.252
Router-DC(config-if)# clock rate 64000
Router-DC(config-if)# no shutdown

%LINK-5-CHANGED: Interface Serial0/2/0, changed state to down
Router-DC(config-if)#end
Router-DC#wr
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

%SYS-5-CONFIG_I: Configured from console by console

Building configuration...
[OK]
```

2. Router-JKT

```
Router>enable
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname Router-JKT
Router-JKT(config)#int Gi0/0
Router-JKT(config-if)# ip address 192.168.10.1 255.255.255.0
Router-JKT(config-if)# no shutdown

Router-JKT(config-if)#int Se0/3/0
Router-JKT(config-if)# ip address 10.0.12.2 255.255.255.252
Router-JKT(config-if)# no shutdown

Router-JKT(config-if)#end
Router-JKT#wr
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

%LINK-5-CHANGED: Interface Serial0/3/0, changed state to up

%SYS-5-CONFIG_I: Configured from console by console

Building configuration...
[OK]
```

3. Router-BDG

```
Router>enable
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname Router-BDG
Router-BDG(config)#int Gi0/0
Router-BDG(config-if)# ip address 192.168.20.1 255.255.255.0
Router-BDG(config-if)# no shutdown

Router-BDG(config-if)#int Se0/3/0
Router-BDG(config-if)# ip address 10.0.13.2 255.255.255.252
Router-BDG(config-if)# no shutdown

Router-BDG(config-if)#end
Router-BDG#wr
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

%LINK-5-CHANGED: Interface Serial0/3/0, changed state to up

%SYS-5-CONFIG_I: Configured from console by console

Building configuration...
[OK]
```

4. Router-SBY

```
Router>enable
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname Router-SBY
Router-SBY(config)#int Gi0/0
Router-SBY(config-if)# ip address 192.168.30.1 255.255.255.0
Router-SBY(config-if)# no shutdown

Router-SBY(config-if)#int Se0/3/0
Router-SBY(config-if)# ip address 10.0.14.2 255.255.255.252
Router-SBY(config-if)# no shutdown

Router-SBY(config-if)#end
Router-SBY#wr
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

%LINK-5-CHANGED: Interface Serial0/3/0, changed state to up

%SYS-5-CONFIG I: Configured from console by console
```

Step 5 – enable OSPF routing untuk semua Router

Berikut merupakan konfigurasi CLI tiap router:

1. Router-DC

```
Router-DC#enable
Router-DC#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router-DC(config)#router ospf 1
Router-DC(config-router)# network 192.168.100.0 0.0.0.255 area 0
Router-DC(config-router)# network 10.0.12.0 0.0.0.3 area 0
Router-DC(config-router)# network 10.0.13.0 0.0.0.3 area 0
Router-DC(config-router)# network 10.0.14.0 0.0.0.3 area 0
Router-DC(config-router)#end
Router-DC#wr
%SYS-5-CONFIG_I: Configured from console by console

Building configuration...
[OK]
```

2. Router-JKT

```
Router-JKT#enable
Router-JKT#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router-JKT(config)#router ospf 1
Router-JKT(config-router)# network 192.168.10.0 0.0.0.255 area 0
Router-JKT(config-router)# network 10.0.12.0 0.0.0.3 area 0
Router-JKT(config-router)#end
Router-JKT#wr
%SYS-5-CONFIG_I: Configured from console by console

00:22:50: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.100.1 on Serial0/3/0 from LOADING to FULL, Loading Done

Building configuration...
[OK]
```

3. Router-BDG

```
Router-BDG#enable
Router-BDG#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router-BDG(config)#router ospf 1
Router-BDG(config-router)# network 192.168.20.0 0.0.0.255 area 0
Router-BDG(config-router)# network 10.0.13.0 0.0.0.3 area 0
Router-BDG(config-router)#end
Router-BDG#wr
%SYS-5-CONFIG_I: Configured from console by console
```

```
Building configuration...
[OK]
```

4. Router-SBY

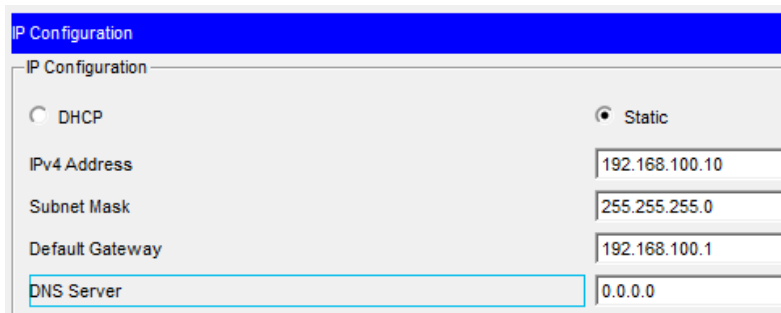
```
Router-SBY#enable
Router-SBY#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router-SBY(config)#router ospf 1
Router-SBY(config-router)# network 192.168.30.0 0.0.0.255 area 0
Router-SBY(config-router)# network 10.0.14.0 0.0.0.3 area 0
Router-SBY(config-router)#end
Router-SBY#wr
%SYS-5-CONFIG_I: Configured from console by console
```

```
Building configuration...
[OK]
```

```
Router-SBY#
00:23:32: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.100.1 on Serial0/3/0 from LOADING to FULL, Loading Done
```

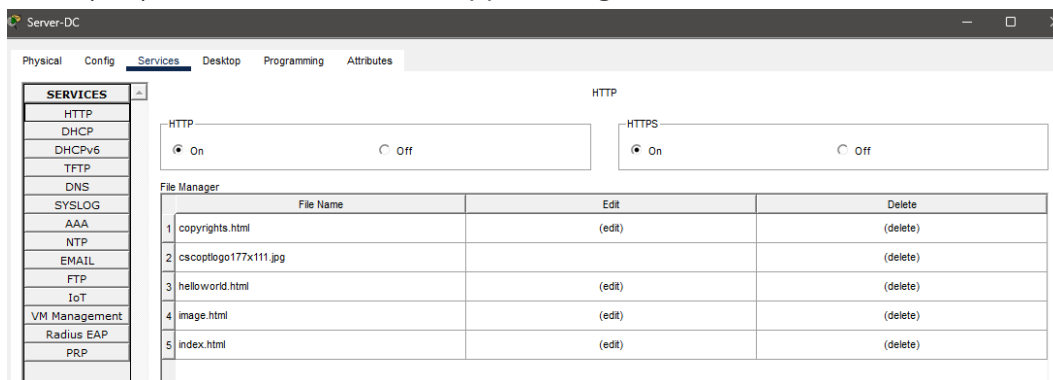
Step 6 – konfigurasi Server-DC

Konfigurasi IP Server-DC



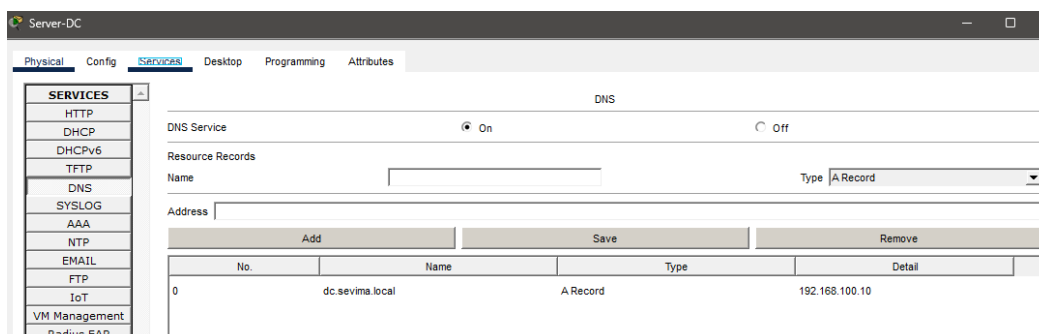
Enable service Server-DC

1. HTTP (ON) – untuk simulasi web app hosting



No.	File Name	Edit	Delete
1	copyrights.html	(edit)	(delete)
2	cscoplogo177x111.jpg		(delete)
3	helloworld.html	(edit)	(delete)
4	image.html	(edit)	(delete)
5	index.html	(edit)	(delete)

2. DNS – untuk name resolution IP



No.	Name	Type	Detail
0	dc.sevima.local	A Record	192.168.100.10

Step 7 – setup IP PC

PC-JKT1

IP Configuration	
Interface	FastEthernet0
IP Configuration	
☐ DHCP	☒ Static
IPv4 Address	192.168.10.10
Subnet Mask	255.255.255.0
Default Gateway	192.168.10.1
DNS Server	0.0.0.0

PC-JKT2

IP Configuration	
Interface	FastEthernet0
IP Configuration	
☐ DHCP	☒ Static
IPv4 Address	192.168.10.11
Subnet Mask	255.255.255.0
Default Gateway	192.168.10.1
DNS Server	0.0.0.0

PC-BDG1

IP Configuration	
Interface	FastEthernet0
IP Configuration	
☐ DHCP	☒ Static
IPv4 Address	192.168.20.10
Subnet Mask	255.255.255.0
Default Gateway	192.168.20.1
DNS Server	0.0.0.0

PC-BDG2

IP Configuration	
Interface	FastEthernet0
IP Configuration	
☐ DHCP	☒ Static
IPv4 Address	192.168.20.11
Subnet Mask	255.255.255.0
Default Gateway	192.168.20.1
DNS Server	0.0.0.0

PC-SBY1

IP Configuration	
Interface	FastEthernet0
IP Configuration	
☐ DHCP	☒ Static
IPv4 Address	192.168.30.10
Subnet Mask	255.255.255.0
Default Gateway	192.168.30.1
DNS Server	0.0.0.0

PC-SBY2

IP Configuration	
Interface	FastEthernet0
IP Configuration	
☐ DHCP	☒ Static
IPv4 Address	192.168.30.11
Subnet Mask	255.255.255.0
Default Gateway	192.168.30.1
DNS Server	0.0.0.0

Verifikasi

Cek ping di PC-JKT1 Command Prompt

```
C:\>ping 192.168.20.10

Pinging 192.168.20.10 with 32 bytes of data:

Reply from 192.168.20.10: bytes=32 time=15ms TTL=125
Reply from 192.168.20.10: bytes=32 time=2ms TTL=125
Reply from 192.168.20.10: bytes=32 time=10ms TTL=125
Reply from 192.168.20.10: bytes=32 time=2ms TTL=125

Ping statistics for 192.168.20.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 15ms, Average = 7ms

C:\>ping 192.168.30.10

Pinging 192.168.30.10 with 32 bytes of data:

Reply from 192.168.30.10: bytes=32 time=15ms TTL=125
Reply from 192.168.30.10: bytes=32 time=2ms TTL=125
Reply from 192.168.30.10: bytes=32 time=2ms TTL=125
Reply from 192.168.30.10: bytes=32 time=2ms TTL=125

Ping statistics for 192.168.30.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 15ms, Average = 5ms

C:\>ping 192.168.100.10

Pinging 192.168.100.10 with 32 bytes of data:

Reply from 192.168.100.10: bytes=32 time=14ms TTL=126
Reply from 192.168.100.10: bytes=32 time=2ms TTL=126
Reply from 192.168.100.10: bytes=32 time=1ms TTL=126
Reply from 192.168.100.10: bytes=32 time=1ms TTL=126

Ping statistics for 192.168.100.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 14ms, Average = 4ms
```

Cek ping di PC-SBY2 Command Prompt

```
C:\>ping 192.168.10.10

Pinging 192.168.10.10 with 32 bytes of data:

Reply from 192.168.10.10: bytes=32 time=2ms TTL=125
Reply from 192.168.10.10: bytes=32 time=2ms TTL=125
Reply from 192.168.10.10: bytes=32 time=2ms TTL=125
Reply from 192.168.10.10: bytes=32 time=2ms TTL=125

Ping statistics for 192.168.10.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 2ms, Average = 2ms

C:\>ping 192.168.20.10

Pinging 192.168.20.10 with 32 bytes of data:

Reply from 192.168.20.10: bytes=32 time=2ms TTL=125
Reply from 192.168.20.10: bytes=32 time=2ms TTL=125
Reply from 192.168.20.10: bytes=32 time=4ms TTL=125
Reply from 192.168.20.10: bytes=32 time=3ms TTL=125

Ping statistics for 192.168.20.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 4ms, Average = 2ms

C:\>ping 192.168.100.10

Pinging 192.168.100.10 with 32 bytes of data:

Reply from 192.168.100.10: bytes=32 time=1ms TTL=126
Reply from 192.168.100.10: bytes=32 time=1ms TTL=126
Reply from 192.168.100.10: bytes=32 time=1ms TTL=126
Reply from 192.168.100.10: bytes=32 time=1ms TTL=126

Ping statistics for 192.168.100.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 1ms, Average = 1ms
```

Cek ping di PC-BDG1 Command Prompt

```
C:\>ping 192.168.10.11

Pinging 192.168.10.11 with 32 bytes of data:

Reply from 192.168.10.11: bytes=32 time=2ms TTL=125
Reply from 192.168.10.11: bytes=32 time=18ms TTL=125
Reply from 192.168.10.11: bytes=32 time=2ms TTL=125
Reply from 192.168.10.11: bytes=32 time=2ms TTL=125

Ping statistics for 192.168.10.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 18ms, Average = 6ms

C:\>ping 192.168.30.11

Pinging 192.168.30.11 with 32 bytes of data:

Reply from 192.168.30.11: bytes=32 time=15ms TTL=125
Reply from 192.168.30.11: bytes=32 time=11ms TTL=125
Reply from 192.168.30.11: bytes=32 time=11ms TTL=125
Reply from 192.168.30.11: bytes=32 time=9ms TTL=125

Ping statistics for 192.168.30.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 9ms, Maximum = 15ms, Average = 11ms

C:\>ping 192.168.100.10

Pinging 192.168.100.10 with 32 bytes of data:

Reply from 192.168.100.10: bytes=32 time=1ms TTL=126
Reply from 192.168.100.10: bytes=32 time=1ms TTL=126
Reply from 192.168.100.10: bytes=32 time=2ms TTL=126
Reply from 192.168.100.10: bytes=32 time=1ms TTL=126

Ping statistics for 192.168.100.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 2ms, Average = 1ms
```